THREE EQUIVALENT CHARACTERIZATIONS OF A UNIVERSALLY INJECTIVE MORPHISM OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is an equivalence cycle (1) -> (2) -> (1) -> (3) -> (2); all four implications are written out, and the final implication also closes the equivalence with (1) through the already proven (2) -> (1).

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Points of a fibre product of algebraic spaces and their field-valued representatives. Properties of Spaces, Lemma spaces-properties-lemma-points-cartesian, used at proof lines 3746-3748. Produces a point of X x_Y X with prescribed projections.
- The criterion for surjectivity of a morphism of algebraic spaces on points. Properties of Spaces, Lemma spaces-properties-lemma-characterize-surjective, used at proof lines 3749-3752 and 3761-3763. Converts pointwise statements into surjectivity of the diagonal and conversely.
- The fpqc sheaf property for morphisms into an algebraic space. Properties of Spaces, Proposition spaces-properties-proposition-sheaf-fpqc, used at proof lines 3739-3742. Concludes that two morphisms Spec(K) -> X that agree after a field extension are equal.
- The identification of points in a fibre product via a common field extension. used at proof lines 3693-3699 and 3721-3726. Formally unwinds what injectivity of |Y' x_Y X| -> |Y'| means.
- Stability of the condition on the diagonal under base change. used at proof lines 3754-3755. Deduces condition (2) for arbitrary Y' from the case Y' = Y.

Source reference. Original lemma 03H5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4554–4554. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03H4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4553–4553. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original proposition 0APL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13479–13479. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. The following are equivalent:*

1. *the map $X(K) \to Y(K)$ is injective for every field K over S*
2. *for every morphism $Y' \to Y$ of algebraic spaces over S the induced map $|Y' \times_Y X| \to |Y'|$ is injective, and*
3. *the diagonal morphism $X \to X \times_Y X$ is surjective.*

Proof. Assume (1). Let $g : Y' \to Y$ be a morphism of algebraic spaces, and denote $f' : Y' \times_Y X \to Y'$ the base change of f. Let K_i, $i = 1, 2$ be fields and let $\varphi_i : \mathrm{Spec}(K_i) \to Y' \times_Y X$ be morphisms such that $f' \circ \varphi_1$ and $f' \circ \varphi_2$ define the same element of $|Y'|$. By definition this means there exists a field Ω and embeddings $\alpha_i : K_i \subset \Omega$ such that the two morphisms $f' \circ \varphi_i \circ \alpha_i : \mathrm{Spec}(\Omega) \to Y'$ are equal. Here is the corresponding commutative diagram

$$\begin{array}{ccccc}
\mathrm{Spec}(\Omega) & \xrightarrow[\alpha_2]{} & \mathrm{Spec}(K_2) & & \\
 & \searrow^{\alpha_1} & & \searrow^{\varphi_2} & \\
 & & \mathrm{Spec}(K_1) & \xrightarrow{\varphi_1} & Y' \times_Y X \xrightarrow{g'} X \\
 & & & & \downarrow{\scriptstyle f'} \qquad \downarrow{\scriptstyle f} \\
 & \searrow & & & Y' \xrightarrow{g} Y.
\end{array}$$

In particular the compositions $g \circ f' \circ \varphi_i \circ \alpha_i$ are equal. By assumption (1) this implies that the morphism $g' \circ \varphi_i \circ \alpha_i$ are equal, where $g' : Y' \times_Y X \to X$ is the projection. By the universal property of the fibre product we conclude that the morphisms $\varphi_i \circ \alpha_i : \mathrm{Spec}(\Omega) \to Y' \times_Y X$ are equal. In other words φ_1 and φ_2 define the same point of $Y' \times_Y X$. We conclude that (2) holds.

Assume (2). Let K be a field over S, and let $a, b : \mathrm{Spec}(K) \to X$ be two morphisms such that $f \circ a = f \circ b$. Denote $c : \mathrm{Spec}(K) \to Y$ the common value. By assumption $|\mathrm{Spec}(K) \times_{c,Y} X| \to |\mathrm{Spec}(K)|$ is injective. This means there exists a field Ω and embeddings $\alpha_i : K \to \Omega$ such that

$$\begin{array}{ccc}
\mathrm{Spec}(\Omega) & \xrightarrow[\alpha_1]{} & \mathrm{Spec}(K) \\
{\scriptstyle \alpha_2}\downarrow & & \downarrow{\scriptstyle a} \\
\mathrm{Spec}(K) & \xrightarrow{b} & \mathrm{Spec}(K) \times_{c,Y} X
\end{array}$$

is commutative. Composing with the projection to $\mathrm{Spec}(K)$ we see that $\alpha_1 = \alpha_2$. Denote the common value α. Then we see that $\{\alpha : \mathrm{Spec}(\Omega) \to \mathrm{Spec}(K)\}$ is a fpqc covering of $\mathrm{Spec}(K)$ such that the two morphisms a, b become equal on the members of the covering. By Properties of Spaces, Proposition 0APL we conclude that $a = b$. We conclude that (1) holds.

Assume (3). Let $x, x' \in |X|$ be a pair of points such that $f(x) = f(x')$ in $|Y|$. By Properties of Spaces, Lemma 03H4 we see there exists a $x'' \in |X \times_Y X|$ whose projections are x and x'. By assumption and Properties of Spaces, Lemma 03H5

there exists a $x''' \in |X|$ with $\Delta_{X/Y}(x''') = x''$. Thus $x = x'$. In other words f is injective. Since condition (3) is stable under base change we see that f satisfies (2).

Assume (2). Then in particular $|X \times_Y X| \to |X|$ is injective which implies immediately that $|\Delta_{X/Y}| : |X| \to |X \times_Y X|$ is surjective, which implies that $\Delta_{X/Y}$ is surjective by Properties of Spaces, Lemma 03H5. □